

A 5-Level Inverter Scheme Using Single DC Link With Reduced Number of Floating Capacitors and Switches for Open-End IM Drives

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Abstract—This paper presents a 5-level inverter topology for open-end induction motor drives by using a single dc source. The open stator windings of the drive are supplied with a 3-level flying capacitor inverter from one end and capacitor-fed 2-level inverter from another end. The voltage ratio of the dc link to the capacitor in 2-level inverter is maintained at 4:1 ratio to generate five-level voltage output. The capacitor in 2-level inverter is balanced by the switching redundant vector combinations from both the inverters while the floating capacitors in the 3-level inverter are balanced by using redundant switching states. The proposed topology gives 5-level operation with less number of floating capacitors and power semiconductor switches compared to other existing topologies. Also, the balancing of the capacitors is independent of load power factor and modulation index. Further, the generalization of the proposed dual inverter scheme for any n -level inverter is also included in this paper. The experimental results and required analysis are also presented to validate the inverter scheme.

Index Terms—Flying capacitor (FC), open-end induction motor (OEIM) drive, pulsewidth modulation (PWM), space vector (SV), 5-level multilevel inverter (MLI).

I. INTRODUCTION

MULTILEVEL inverters (MLI) are the most suitable alternative for high-power motor drive applications. It offers several advantages over the 2-level inverter, like low voltage stress across the switches, smoother phase voltage output, lower dv/dt across the motor phase winding, ripple free shaft torque, etc. Due to its attractive features, extensive research has been

going on for a few decades on multilevel converters. Different multilevel topologies have been registered in literatures; among them neutral point clamp (NPC) inverter, flying capacitor (FC) inverter, and cascaded H-bridge (CHB) are the most common [1]. The major application field of the MLI are induction motor drives and grid tied operation.

The NPC topology [2] requires additional numbers of clamping diodes with increase in output voltage levels. Also, it has inherent neutral point voltage balancing problem, which depends on the modulation index, load current, and fundamental frequency [3], [4]. The FC inverter requires additional numbers of floating capacitors to generate multilevel output voltage using a single dc link. Also, special care is required to balance all these capacitors in each PWM cycle [5]. While the CHB topology is free from the abovementioned demerits, it requires more number of isolated dc sources, which makes the total system bulky [6]. To reduce the component count, voltage stress across the switches and switching loss in an inverter scheme, hybrid and derivative topologies have been established by combining basic MLI structures [7]–[10].

Apart from traditional single-fed inverters, i.e., feeding drives from one side, dual-fed inverters have also been reported in literature for open-end induction motor (OEIM) drives. Dual-fed inverters are more reliable compared to single-fed inverters [11], [12]. Under circumstances of failure of one inverter, other inverter can still be in operation with reduced number of levels. Among these, a few basic dual-fed inverters use two isolated supplies to generate multilevel output voltage [13]–[15], which makes the total system bulky. Dual-fed inverter schemes proposed in [16]–[18] use single dc source, where either inverter operates with common-mode voltage being eliminated or by applying same common-mode voltages from both the inverters. But these schemes reduce the number of voltage levels with a 15% reduction in output phase voltage compared to dual-inverter schemes supplied from two isolated dc links. The dual-inverter scheme proposed in [19] uses a single dc source, but obtains only a 3-level voltage space vector (SV) structure.

This paper presents a 5-level dual-fed inverter scheme for OEIM drives using a single dc source. This topology aims to eliminate the usage of bulky transformers required for an isolated dc source in dual-inverter topologies. At the same time, it

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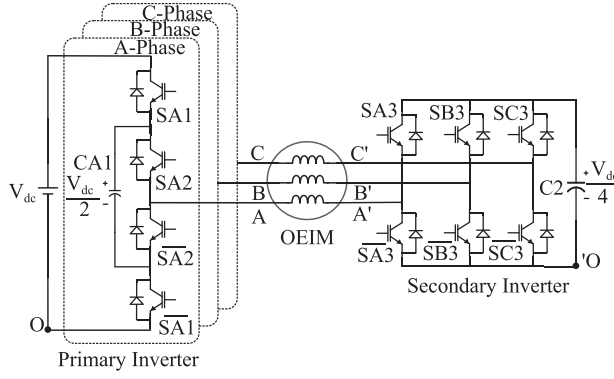


Fig. 1. Power circuit diagram of the proposed 5-level inverter scheme.

can generate fundamental phase voltage amplitude up to 0.577 times of the dc-link voltage by modulating SV-based PWM technique. This topology uses a single dc source, which is connected to the 3-level inverter while the 2-level inverter is supplied with a capacitor. All the capacitors are maintained at their nominal voltage value by selecting proper switching state for the entire modulation range. The capacitor can be charged by using phase current, which eliminates the need of additional precharging circuit. Switching loss is also evaluated analytically and comparison with a few basic and hybrid topologies are also shown. Generalization of the proposed topology to extend up to any n level is included. The hardware experimental results for closed-loop rotor field-oriented control (FOC) and open-loop V/f control during steady and transient state are included. Further, the fault tolerance capability of dual inverter scheme is also included.

II. PROPOSED INVERTER TOPOLOGY

The proposed 5-level inverter scheme having minimum components count is shown in Fig. 1 along with OEIM drive. The drive is powered by a 3-level FC inverter from one end as primary inverter and a 2-level inverter from another end as secondary inverter. In this scheme, the primary inverter is fed with a dc source of voltage rating V_{dc} and the secondary inverter is fed with a floating capacitor, whose voltage is maintained at $V_{dc}/4$. The floating capacitors in the primary inverter are maintained at a voltage of $V_{dc}/2$. The pole voltages of primary inverter (V_{AO} , V_{BO} , V_{CO}) can take values of 0, $V_{dc}/2$, V_{dc} . The secondary inverter pole voltages ($V_{A'O'}$, $V_{B'O'}$, $V_{C'O'}$) can take values 0 or $V_{dc}/4$. The phase voltage across the induction machine winding is governed by the following:

$$V_{AA'} = V_{AO} - V_{A'O'} + V_{OO'} \quad (1)$$

$$V_{BB'} = V_{BO} - V_{B'O'} + V_{OO'} \quad (2)$$

$$V_{CC'} = V_{CO} - V_{C'O'} + V_{OO'} \quad (3)$$

In the abovementioned equations, $V_{OO'}$ is the common mode voltage. As the point O and O' are isolated from each other, there is no path for common mode current to flow [15]. The

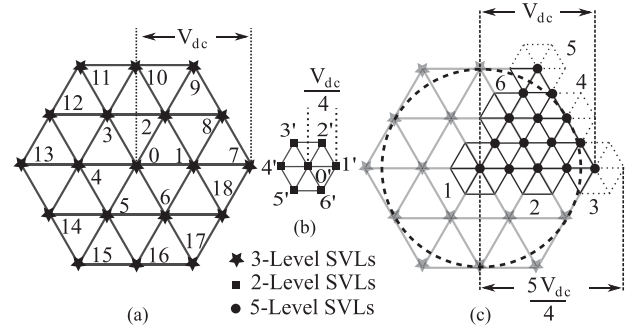


Fig. 2. Generation of 5-level SV structure. (a) Three-level SV structure. (b) Two-level SV structure. (c) Five-level SV structure.

voltage stress across each switch in the primary inverter during OFF state is $V_{dc}/2$ and that of the secondary inverter is $V_{dc}/4$. In Fig. 1, $\bar{S}_{xn}$ and S_{xn} are the complimentary pair of switches, where “x” is A, B, or C and “n” is 1, 2, or 3.

III. PRINCIPLE OF OPERATION

A. Formation of SV Structure

In the proposed inverter scheme, the output of the primary inverter will generate a 3-level SV structure of radius V_{dc} and secondary inverter will generate a 2-level SV structure of radius $V_{dc}/4$ as shown in Fig. 2. The output voltage of the primary inverter in SV domain is given by

$$\vec{V}_1 = V_{AO} \angle 120^\circ + V_{BO} \angle 240^\circ + V_{CO} \angle 0^\circ \quad (4)$$

whereas, for secondary inverter, it is defined as

$$\vec{V}_2 = V_{A'O'} \angle 180^\circ + V_{B'O'} \angle 300^\circ + V_{C'O'} \angle 60^\circ \quad (5)$$

The 5-level SV structure is generated by superimposing the 2-level SV structure with the 3-level SV structure. The effective voltage across motor phase $\vec{V}$ winding is given by $\vec{V}_1 + \vec{V}_2$, i.e.,

$$\vec{V} = V_{AA'} \angle 120^\circ + V_{BB'} \angle 240^\circ + V_{CC'} \angle 0^\circ \quad (6)$$

Fig. 2(c) shows the construction of 5-level SV structure by superimposing 2-level SVs of radius $V_{dc}/4$ over the 3-level SV structure of radius V_{dc} for 60 of the hexagon. Here, we cannot generate reference vector beyond the circle, i.e., magnitude of the reference can have a maximum value $0.866 V_{dc}$ beyond which the secondary capacitor voltage cannot be balanced. The reason is discussed in Section B.

A total of 103 unique vector combinations are possible from secondary and primary side for generating total 61 SV locations (SVLs) in 5-level SV structure. Fig. 3 and Table I show the formation of 5-Level SVs by the combinations of primary and secondary SVs for 60 region of the 5-level SV structure in detail. Fig. 3 shows that the SVLs marked with dot have two possible combinations, while SVLs marked with star have only one possible combination (where zero vector applied from secondary inverter side).

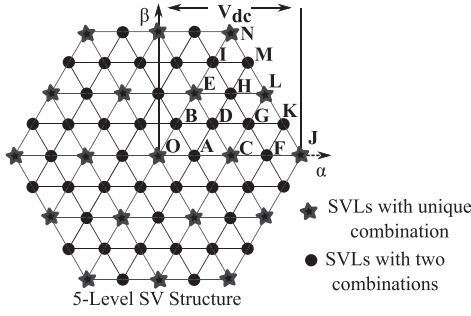


Fig. 3. Combinations of SV from primary and secondary inverter for generation of each SVL in 5-level SV structure.

TABLE I
GENERATION OF 5-LEVEL SVs BY COMBINATION OF PRIMARY AND SECONDARY INVERTER SVs

Combined 5-Level SV	SV from Primary Inverter	SV from Secondary Inverter
OA	01	0'1'
	0	0'4'
OB	02	0'2'
	0	0'5'
OC	01	0'
	01	0'6'
OD	02	0'3'
	02	0'
OE	01	0'4'
	07	0'1'
OF	01	0'5'
	08	0'2'
OG	02	0'4'
	08	0'1'
OH	09	0'2'
	02	0'5'
OI	07	0'
	08	0'3'
OK	07	0'6'
	08	0'
OL	08	0'6'
	09	0'3'
OM	09	0'
	09	0'

B. Capacitor Voltage Balancing

To generate reference voltage, vector nearest possible SVs from 5-level SV structure have to switch. Now, for each selected SV from 5-level SV structure, there are two possible combinations (except star marked points, Fig. 3) of SV from primary and secondary side. In order to balance all the capacitor voltages, proper vector combination have to select from primary and secondary side inverter corresponding to each selected SV from 5-level structure. Hence, to generate switched average reference voltage vector, considering the current direction in three phases and the secondary inverter capacitor voltage, first, secondary SV will be chosen and then corresponding primary SV will be selected. Current through capacitor (C2) in secondary inverter is governed by the following:

$$I_{C2} = I_a * f(SA3) + I_b * f(SB3) + I_c * f(SC3) \quad (7)$$

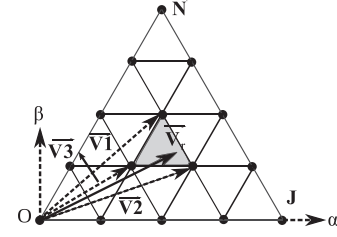


Fig. 4. Generation of reference voltage vector by equivalent 5-level SV of the dual-inverter system.

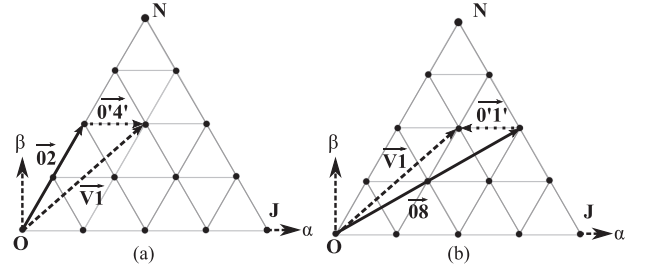


Fig. 5. Synthesis of 5-level SV $\vec{V}_1$ by primary and secondary SV as $(\vec{02} + \vec{0'4'})$ or as $(\vec{08} + \vec{0'1'})$.

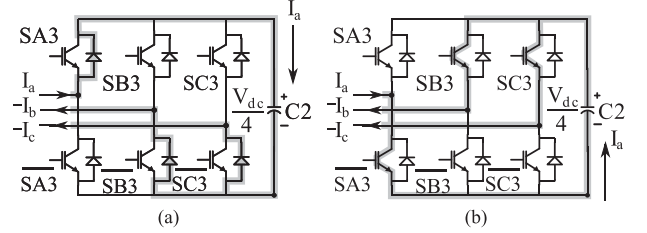


Fig. 6. (a) Secondary floating capacitor is charging for switching of SV $\vec{0'1'}$. (b) Secondary floating capacitor is discharging for switching of SV $\vec{0'4'}$.

where $f(SA3)$, $f(SB3)$, $f(SC3)$ are the switching function for the secondary inverter switches SA3, SB3, SC3, and can take values $f(Sx3) = 1$ when switch $Sx3$ is ON and $f(Sx3) = 0$ when switch $Sx3$ is OFF and "x" is A, B, or C.

It can be seen from Fig. 4 that to generate the reference vector $\vec{V}_r$, nearest SVs $\vec{V}_1$, $\vec{V}_2$, and $\vec{V}_3$ have to switch from equivalent 5-level SV structure. Now, available SV combinations from primary and secondary inverter for composing equivalent 5-Level SV $\vec{V}_1$ are 1) $\vec{02}$ from primary side and $\vec{0'4'}$ from secondary side [see Fig. 5(a)] or 2) $\vec{08}$ from primary side and $\vec{0'1'}$ from secondary side [see Fig. 5(b)] and corresponding switching state in the secondary inverter for SV $\vec{0'1'}$ and $\vec{0'4'}$ are shown in Fig. 6. Likewise, 5-level SV $\vec{V}_2$ and $\vec{V}_3$ can also be composed in two possible ways by applying proper SV combination from primary and secondary inverter side.

Let us assume any arbitrary current direction in three phases like, I_a is positive and I_b , I_c are negative, and also consider

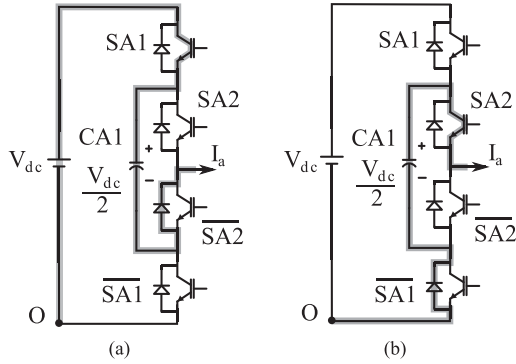


Fig. 7. Switching state for pole voltage $V_{AO} = V_{dc}/2$. (a) Primary inverter floating capacitor is charging and (b) primary inverter floating capacitor is discharging with the positive phase current I_a .

that the secondary capacitor voltage is below the nominal value. For the abovementioned condition, during the instant when SV $\vec{V}_1$ need to switch for generating $\vec{V}_r$, vector $\vec{0'1'}$ from secondary side should be switched to charge the capacitor [see (7)] and corresponding primary vector $\vec{08}$ will be switched. Similarly, with the abovementioned current direction, if the secondary capacitor voltage is above the nominal value, then vector $\vec{0'4'}$ will be switched from secondary side to discharge the secondary capacitor along with vector $\vec{02}$ from primary side. For the considered current direction, the secondary capacitor is charging with phase current I_a when secondary vector $\vec{0'1'}$ is switched [see Fig. 6(a)] and the secondary capacitor is discharging by phase current I_a with the switching of vector $\vec{0'4'}$ [see Fig. 6(b)]. In 5-level SV structure (see Fig. 3), only one SV combination is available for SVLs marked in star, but at those locations, zero vector is switched from secondary inverter. Hence, the secondary side capacitor will be unaffected. So, it is possible to balance the secondary capacitor in every SVLs in 5-level SV structure by proposed switching strategy.

If magnitude of the reference voltage vector goes beyond the dotted circle (see Fig. 2), then we cannot charge back the secondary floating capacitor as there is only one vector combination available from primary and secondary inverter. At those points, active SV will be applied from secondary inverter with the corresponding SV from primary inverter and hence, secondary capacitor will be discharged and due to lack in charging states (i.e., vector redundancies) we cannot charge it back. So, linear operation zone of the inverter scheme is limited up to the dotted circle as shown in Fig. 2.

Primary side floating capacitors will be affected only when pole voltage $V_{dc}/2$ is applied from the primary inverter. Considering positive direction of current (see Fig. 7), the floating capacitor can be either charged by applying switching redundancy ($V_{dc} - V_{dc}/2$) or discharged by applying $(0 + V_{dc}/2)$ while generating a pole voltage of $V_{dc}/2$. This capacitor current can be written as

$$I_{Cx1} = f(Sx1) * I_x - f(Sx2) * I_x \quad (8)$$

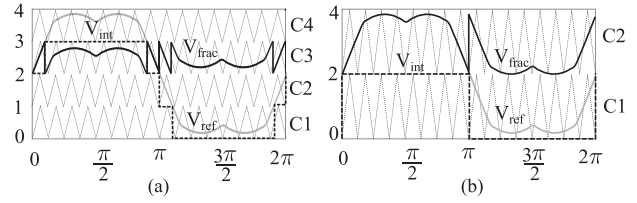


Fig. 8. Modulation technique. (a) During normal operation, V_{ref} is the reference signal, C1–C4 are the four carriers, V_{int} is the integer component, and V_{frac} is the fractional component of reference voltage. (b) During fault condition, V_{ref} is the reference phase voltage, C1 and C2 are the two carriers, V_{int} is the integer component, and V_{frac} is the fractional component of reference voltage.

where $f(Sx1)$ and $f(Sx2)$ are the switching functions for the primary inverter switches $Sx1$ and $Sx2$ and “x” is A, B, or C and I_x is the phase current.

C. Modulation Technique

The proposed inverter scheme uses level-shifted carrier-based pulsewidth modulation (LSC-PWM) for modulating its switches [20]. In this case, four level shifted carrier signal are used to compare with the modulating signal. Here, modulating signal (V_r) is scaled between 0 to 4 as

$$V_{ref} = 2 + (V_r * 2) / V_{dc}. \quad (9)$$

Then it is compared with the carrier signals. However, reference signal is decomposed to integer and fractional part as

$$V_{ref} = V_{int} + V_{frac}. \quad (10)$$

V_{int} is the nearest integer less than or equal to V_{ref} , and it can take values of 0, 1, 2, 3, and V_{frac} lies in the interval [0,1] [see Fig. 8(a)]. This V_{int} and V_{frac} information from all the three phases gives a particular SV from equivalent 5-level SV structure and the time duration for which that SV will be switched. Now, depending upon the current direction and secondary capacitor voltage, the 5-level SV will be synthesized by 2-level and 3-level SV generated by secondary and primary inverter, respectively.

D. Inverter Operation During Fault

In the proposed inverter scheme, the OEIM drive is fed by two inverters, namely, primary inverter and secondary inverter (see Fig. 1). Primary inverter is supplied from a dc source and it supplies the required active power for the drive operation. The secondary inverter is capacitor-fed inverter, which is used to generate more of number of voltage levels. In case of power switch failure in secondary inverter, the drive can still be in operation with secondary inverter bypassed (zero vector will be applied from the secondary inverter). As the dc source is connected with the primary inverter, the drive can be operated at full modulation index and rated power during the fault condition. The switch in secondary inverter can be faulted with either being short circuited or open circuited. As there are six switches in secondary inverter, a total of 2^6 conditions are possible for

fault. These possibilities can be broadly classified into four categories [21]:

- 1) single switch short circuit (S.C.) fault;
- 2) single switch open circuit (O.C.) fault;
- 3) phase leg S.C. fault;
- 4) phase leg O.C. fault.

The proposed drive scheme can still be in operation in all the abovementioned fault cases, except during phase leg O.C. fault, because during this fault, the secondary inverter cannot be bypassed. During the fault condition, the LSC-PWM method will be changed from 5-level to 3-level scheme [see Fig. 8(b)] with only two level shifted carrier. Switching pulses are provided only for primary inverter while in secondary inverter, either all the top switches or all the bottom switches are turned ON. During the fault, the primary inverter (3-level FC) applies three pole voltage (V_{AO}) levels of 0, $V_{dc}/2$, and V_{dc} from the phase terminals. Hence, the proposed scheme is more reliable compared to traditional single-fed inverter schemes.

IV. SWITCHING LOSS CALCULATION AND COMPARISON WITH OTHER TOPOLOGIES

The switching loss for the proposed converter is calculated for different frequencies of operation for a three-phase 15-kW, 415-V, 50-Hz OEIM drive. This loss is comprised of the primary and secondary inverter switching loss during turning the power IGBT device ON and OFF. For loss calculation, the dc-bus voltage is taken as 560 V, corresponding to phase voltage rms value 240 V, and peak phase current is taken as 20 A. Switching loss is calculated for SKM100GB12T4 IGBTs and antiparallel diodes at the time of their switching and averaged over the fundamental period. This loss can be calculated by the following:

$$P = \left(\sum_N E_{test} * (I/I_{test})^{(K_i)} * (V/V_{test})^{(K_v)} * \alpha_T \right) * f_{freq} \quad (11)$$

where N is the number of switching instant in one fundamental period, f_{freq} is the fundamental frequency, V is the voltage stress across the device during OFF condition, and I is the current through the switch. E_{test} , I_{test} , V_{test} are the values taken from data sheet, and K_i , K_v , and α_T are the constants given in data sheet. Switches of the primary inverter are switching almost half that of switches of the secondary inverter. Hence switching loss is largely reduced due to lower switching frequency in primary inverter switches, which blocks $V_{dc}/2$ than the switches in secondary inverter, which blocks $V_{dc}/4$. Also switching loss comparison at different frequencies of operation is done analytically with other 5-level topologies like 5-level FC and hybrid 5-level topology [7] and is shown in Fig. 9. During switching loss comparison in all the cases, switching frequencies are chosen such that the capacitor voltage peak-peak ripple remains within 5% of their nominal value with the abovementioned load condition.

Conventional 5-level NPC and FC inverter require extra clamping diodes and floating capacitors, respectively, for their implementation. Also, conventional NPC suffers from neutral

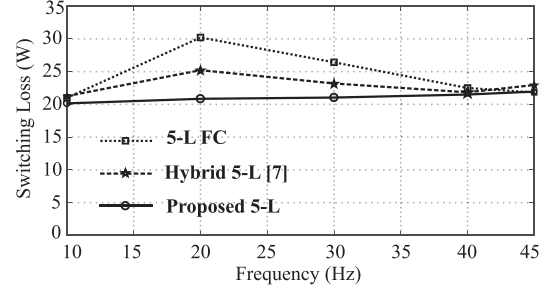


Fig. 9. Switching loss comparison of the proposed topology with FC and hybrid topology.

point unbalancing problem at higher modulation indices [22]. While a CHB inverter requires less amount of components, it requires additional number of isolated dc sources. This topology requires total 18 switches; among them, 12 switches have voltage blocking capability of $V_{dc}/2$ and the rest six switches of $V_{dc}/4$. Capacitors of primary inverter have voltage rating of $V_{dc}/2$, whereas, secondary side capacitor holds only $V_{dc}/4$. Also, all the capacitors are balanced in each PWM cycle irrespective of modulation index and load power factor. The detail comparison in terms of isolated dc source usage, component and switch counts among a few inverter topologies with the proposed topology is shown in Table II.

V. GENERALIZATION OF THE PROPOSED INVERTER SCHEME

The proposed dual inverter scheme can be extended to any number of levels by modifying primary inverter and maintaining secondary 2-level inverter capacitor voltage accordingly. The proposed dual inverter scheme can be generalized for a n -level inverter, where primary inverter will give $(n-2)$ pole voltage levels and the secondary 2-level inverter capacitor voltage should be maintained at $V_{dc}/(2 * (n-3))$ (see Fig. 10) where V_{dc} is the dc-link voltage. The primary inverter capacitors can be balanced by switching redundancies, whereas secondary inverter floating capacitor can be balanced by SV redundancies. With increase in phase voltage level, the nominal value of the secondary capacitor voltage will reduce, which reduces the switching loss in the secondary 2-level inverter. For an instance, to construct a 7-level dual inverter, primary inverter should be a 5-level inverter fed with dc-link voltage V_{dc} and the nominal voltage of the secondary 2-level inverter capacitor will be $V_{dc}/8$.

VI. HARDWARE IMPLEMENTATION

A three-phase 15-kW, 415-V, 50-Hz IM is configured as OEIM after opening its stator neutral terminal. Open-loop V/f and rotor FOC is performed to verify the operation of the proposed inverter scheme. The laboratory prototype is shown in Fig. 11. The proposed topology is controlled by synchronizing TMS320f28335 DSP with Xilinx Spartan3 XCS200 FPGA. Three phase currents and four capacitor voltages are sensed and conveyed to the ADC of the DSP. DSP-generated level

TABLE II
COMPARISON OF 5-LEVEL INVERTER TOPOLOGIES

Topology	Switches			Capacitors		Clamping diodes	DC sources		
	$V_{dc}/2$	$V_{dc}/4$	Total	$V_{dc}/2$	$V_{dc}/4$		V_{dc}	$V_{dc}/2$	$V_{dc}/4$
Proposed (Fig. 3)	12	6	18	3	1	0	1	0	0
Hybrid 5-L [7]	12	12	24	3	3	0	1	0	0
FC 5-L	0	24	24	0	18	0	1	0	0
NPC 5-L	0	24	24	0	0	12	0	0	4
CHB 5-L	0	24	24	0	0	0	0	0	6
Standard ANPC	12	12	24	0	3	0	0	2	0
Hybrid 5-L [8]	0	24	24	0	3	0	0	1	2

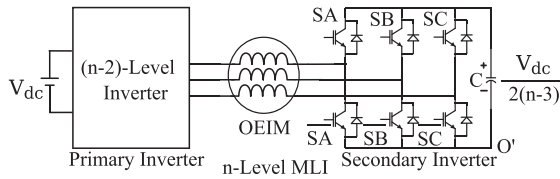


Fig. 10. Generalization of the proposed dual-inverter scheme.

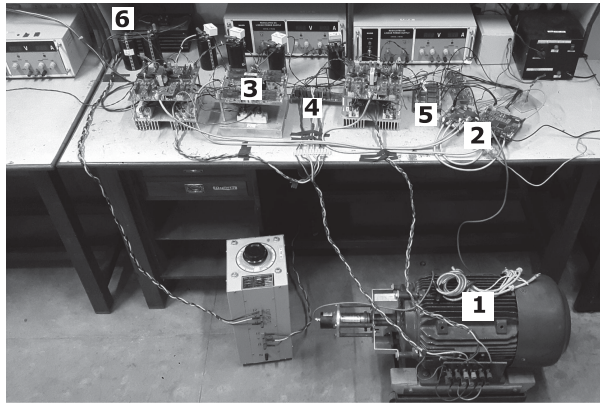


Fig. 11. Laboratory prototype. (a) Induction motor. (b) DSP and FPGA boards. (c) Gate driver boards and IGBTs. (d) Current sensors. (e) Voltage sensors. (f) DC-link generation.

information and PWM pulses, together with current direction and capacitors voltage status, are transferred to FPGA. After decoding this information in FPGAs, switching pulses are generated. Also, $2\text{-}\mu\text{s}$ dead time is provided between the complementary switching pulses. The number of samples are chosen such that the sampling frequency is around 2 kHz. The number of samples are taken from the reference voltage wave varying from 180 samples per fundamental cycle for 10 Hz to 45 samples at 45 Hz fundamental frequency. SKM100GB12T4 IGBTs are used as power switch. Capacitor voltage ripples are maintained within 5% peak to peak of the nominal voltage value. So, capacitor value is designed by $C = (I_p * 2 * T_s) / \Delta V$, which gives secondary capacitor value as 2 mF and primary floating capacitor value as 1 mF, where I_p is peak phase current, ΔV is (peak to peak) capacitor voltage ripple, and T_s is the sampling time. For laboratory prototype, all the capacitors of 2.2 mF are used.

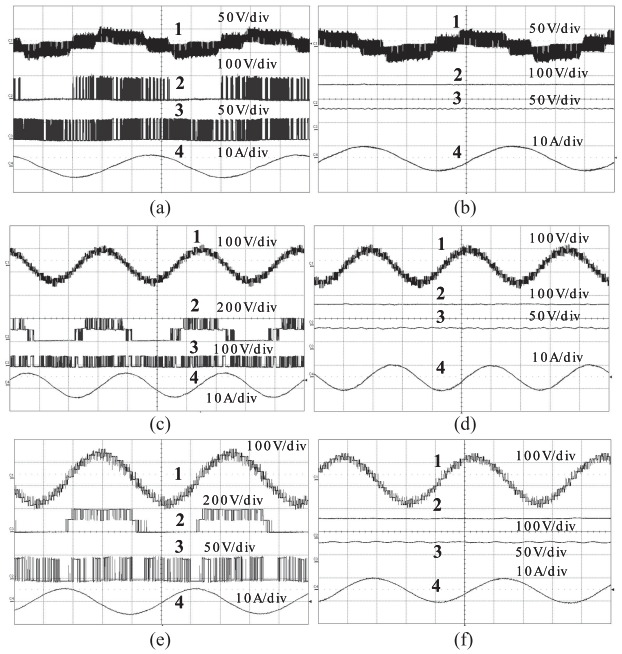


Fig. 12. (a) 10-Hz operation, X-axis: 20 ms/div (1) phase voltage $V_{AA'}$, (2) primary inverter pole voltage V_{AO} , (3) secondary inverter pole voltage $V_{A'O'}$, and (4) phase current I_A . (b) 10-Hz operation, X-axis: 20 ms/div (1) phase voltage $V_{AA'}$, (2) primary inverter floating capacitor voltage V_{CA1} , (3) secondary inverter floating capacitor voltage V_{C2} , and (4) phase current I_A . (c) 30-Hz operation, X-axis: 10 ms/div (1) phase voltage $V_{AA'}$, (2) primary inverter pole voltage V_{AO} , (3) Secondary inverter pole voltage $V_{A'O'}$, and (4) phase current I_A . (d) 30-Hz operation, X-axis: 10 ms/div (1) phase voltage $V_{AA'}$, (2) primary inverter floating capacitor voltage V_{CA1} , (3) secondary inverter floating capacitor voltage V_{C2} , and (4) phase current I_A . (e) 45-Hz operation, X-axis: 5 ms/div (1) phase voltage $V_{AA'}$, (2) primary inverter pole voltage V_{AO} , (3) secondary inverter pole voltage $V_{A'O'}$, and (4) phase current I_A . (f) 45-Hz operation, X-axis: 5 ms/div (1) phase voltage $V_{AA'}$, (2) primary inverter floating capacitor voltage V_{CA1} , (3) secondary inverter floating capacitor voltage V_{C2} , and (4) phase current I_A .

VII. EXPERIMENTAL RESULTS

Experimental results of open-loop V/f control are shown in Fig. 12 for OEIM drives. Fig. 12(a)–(c) are the steady-state results for phase voltage, pole voltages of the primary inverter and the secondary inverter, phase current for 10, 30, and 45 Hz operation, respectively, and Fig. 12(d)–(f) shows the phase voltage,

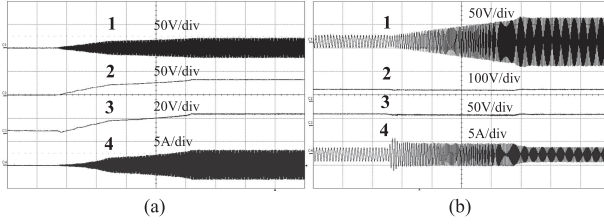


Fig. 13. (a) Starting, X-axis: 2 s/div (1) phase voltage $V_{AA'}$, (2) primary inverter floating capacitor voltage V_{CA1} , (3) secondary inverter floating capacitor voltage V_{C2} , and (4) phase current I_A . (b) Acceleration, X-axis: 1 s/div (1) phase voltage $V_{AA'}$, (2) primary inverter floating capacitor voltage V_{CA1} , (3) secondary inverter floating capacitor voltage V_{C2} , and (4) phase current I_A 10 A/div.

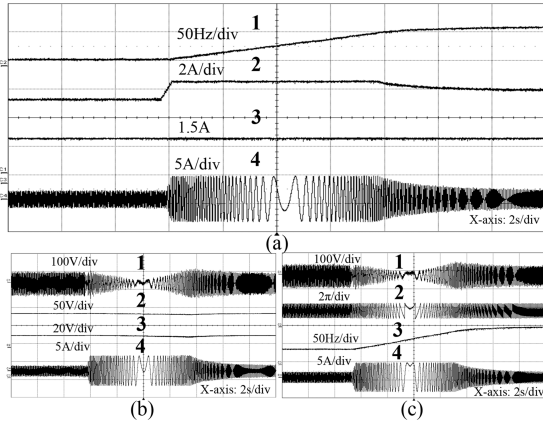


Fig. 14. Speed reversal from -30 to $+30$ Hz under FOC, X-axis: 2 s/div: (a) (1) speed feedback, (2) I_{sq} rotor torque producing component, Y-axis: 2 A/div, (3) I_{sd} rotor magnetization current 1.5 A, and (4) phase current I_A , Y-axis: 5 A/div. (b) (1) Phase voltage Y-axis: 100 V/div, (2) primary inverter floating capacitor voltage, Y-axis: 50 V/div, (3) secondary inverter floating capacitor voltage, Y-axis: 20 V/div, and (4) phase current I_A , Y-axis: 5 A/div. (c) (1) Phase voltage $V_{AA'}$, Y-axis: 100 V/div, (2) rotor flux angle ρ , (3) speed feedback, and (4) phase current I_A , Y-axis: 5 A/div.

primary inverter and secondary inverter floating capacitor voltages, phase current during steady state at abovementioned frequencies, respectively. The abovementioned results are shown for a dc-link voltage of 200 V and at no load of OEIM drive. At 10 Hz, both the inverter pole voltages are only at 2-level operation while in 30 and 45 Hz, primary inverter pole will generate a 3-level voltage output. Fig. 13(a) shows the experimental result during motor starting, and Fig. 13(b) shows the transient result during acceleration from 10 to 40 Hz. abovementioned results show that capacitor voltages are well-controlled in steady state as well as in transient state, and also, the capacitors are charged by the phase currents to their nominal values; hence no additional precharging circuit is required. The dynamic performance of the proposed converter is validated by performing FOC on a 15-kW, 415-V, 50-Hz OEIM drive with a dc-link voltage of 200 V. Fig. 14(a) shows speed feedback I_{sd} , I_{sq} , phase current, and Fig. 14(b) and (c) shows the phase voltage, capacitors voltage, phase current, and ρ during speed reversal from -30 to $+30$ Hz. It is seen that I_{sq} changes instantly to provide

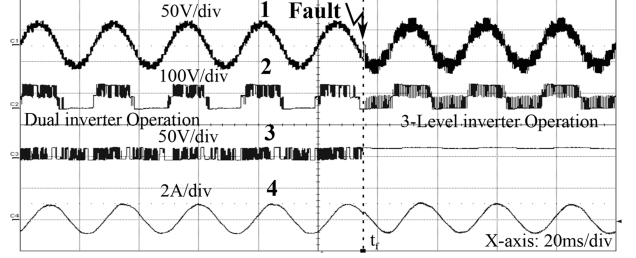


Fig. 15. Demonstration of fault tolerance at 40-Hz operation (1) phase voltage $V_{AA'}$, (2) primary inverter pole voltage V_{AO} , (3) secondary inverter pole voltage $V_{A'O'}$, and (4) phase current I_A .

maximum torque during speed reversal to overcome the inertia, while I_{sd} is maintained at constant value and also capacitor voltages are maintained within desired limit during this speed reversal.

Dual-fed inverter schemes are more reliable compared to traditional single-fed inverter schemes, as discussed in Section III-D. Fault condition due to failure of power switches in secondary inverter is demonstrated with the experimental result in Fig. 15. The fault occurs at time t_f during 40 Hz of operation. Before the fault occurred, both the inverters were driving the OEIM, but after the fault (after t_f), secondary inverter was bypassed and primary inverter alone is driving the OEIM as a 3-level inverter.

VIII. CONCLUSION

This paper proposed a 5-level inverter scheme with reduced components count for OEIM drive. The proposed scheme required only one dc source of voltage V_{dc} and three capacitor of voltages $V_{dc}/2$ for primary FC inverter and only one floating capacitor of voltage $V_{dc}/4$ for secondary 2-level inverter. All the floating capacitor voltages were balanced in each PWM cycle, irrespective of load power factor and modulation index. This feature enabled to use low value of capacitors in medium to high power drive applications. The capacitors could be charged by the phase current during starting; hence, no capacitor precharging circuit was required. Also, single dc source enabled back-to-back power flow. The low voltage switches in the secondary inverter were switching more times compared to high-voltage switches in primary inverter in each fundamental cycle, which reduced switching loss of the overall system. Switching loss for this topology was also compared with some of the existing topologies. Usage of single dc source and low-valued capacitors made it compact and light weight. Further, during fault condition, the converter could be operated as a 3-level inverter up to full modulation index. Generalization of the proposed topology was also included, which could be applied to extend the proposed dual inverter scheme to any n -level inverter. Above-mentioned features make the proposed system more reliable and suitable for application in electric vehicle, industrial motor drives, etc.

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